

M Nikhil

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SUMMARY

Backend & AI Engineer with experience building production LLM applications, AI agents, prompt-engineered workflows, RAG pipelines, and scalable backend systems. Passionate about building reliable agentic AI systems, evaluation pipelines, and developer tools.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript

AI / LLMs: LLMs, Prompt Engineering, AI Agents, Agentic AI, RAG, Vector DBs, Embeddings, LangChain, HuggingFace Transformers, Ollama, llama.cpp, LangGraph

ML: Scikit-learn, XGBoost, Feature Engineering, Predictive Modelling, PEFT, LoRA

Backend: Django, Node.js, FastAPI, REST APIs, OOP, DSA, OS, Debugging, System Design, Agile, SDLC, Celery

DevOps & Tools: Azure Foundry, Azure AI Services, Git, GitHub, Version Control, Docker, Redis, MinIO, Cursor, Claude Code

EXPERIENCE

AI Product Developer Intern

Jan 2026 – July 2026

Roman Technologies Pvt. Ltd., Bangalore, India

- **Voice Agent:** Built and stabilized a production multi-persona AI voice agent with streaming STT → LLM → TTS orchestration, seamless persona switching, and Dinstar/VoBiz telephony integration; reduced first-turn latency by ~52% (2.5s → 1.2s TTFT) through asynchronous cache prewarming and event-loop optimization.
- Improved responsiveness via streaming LLM tokens directly to TTS; resolved production async/streaming failures including audio truncation, STT reconnection loops, interruption deadlocks, and call lifecycle issues; benchmarked open-source TTS models for latency, streaming capability, and deployment readiness.
- **LLM Applications:** Deployed two quantized 7B LLMs on AWS EC2 using llama.cpp; built a production AI chatbot with prompt orchestration, a two-layer guardrail system, and evaluation workflows, reducing unsafe outputs by 90%.
- **CRM Modules:** Applied software engineering principles to design and implement the Accounts module with full CRUD, six view modes (List, Kanban, Grid, Chart, Timeline, Split), 18+ filterable fields, and a deduplication engine for group-based merge resolution.
- Built the Meetings module with Google Meet auto-provisioning, RSVP tracking, and real-time reminder polling; developed the Tasks module with overdue detection, recurring config, and five-status lifecycle management.
- Implemented cross-module Bulk Actions (mass delete, transfer, update) and designed RESTful Django proxy APIs with CSV import/export, soft-delete/restore flows, and timeline audit logging.
- **ERP Modules:** Built RBAC-secured modules with normalized schemas, dynamic dashboards, debounced search, and reusable React UI components using Git-tracked version control workflows.

PROJECTS

AI Video Evidence Management System (AVEMS) | *FastAPI, Celery, Redis, PostgreSQL, Qdrant, Docker*

- Architected the orchestration layer for a multi-model AI video analysis pipeline, designing GPU/CPU task scheduling, shared schemas, and asynchronous workflows enabling independent development across three engineering teams.
- Built the end-to-end ingestion infrastructure using FastAPI, Celery, Redis, PostgreSQL, MinIO, and Qdrant, implementing scalable storage, vector search, and audit pipelines optimized for a single 6GB GPU deployment.

GhostFolio - AI Startup Revival Platform | *React, Gemini API*

- SaaS platform analyzing failed startups and generating AI-powered revival strategies using Gemini API, prompt engineering, and real startup datasets.

errbuddy | *Node.js, CLI, npm* | npmjs.com/package/errbuddy

- Published npm CLI tool intercepting terminal stderr in real time, classifying errors across 11 runtimes and generating contextual AI-powered explanations and suggested fixes.

AI-Based Crime Pattern Prediction | *XGBoost, FastAPI*

- ML system predicting crime patterns on Bengaluru datasets with 87% accuracy; built FastAPI inference pipeline for real-time prediction APIs.

EDUCATION

Sri Sairam College of Engineering, Bangalore, India

2022-2026

B.E. Information Science & Engineering | CGPA: 8.3/10

CERTIFICATIONS

Microsoft Certified: Azure Fundamentals (AZ-900) | Infosys Generative AI Certification | Claude Certified Architect Foundations